

NORTHERN COLORADO, CO, USA

WMO: 724769

Lat: 40.450N

Lon: 105.017W

Elev: 1529

StdP: 84.26

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 94035

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-17.8	-14.7	-22.2	0.6	-16.5	-18.7	0.9	-10.8	13.8	1.4	12.0	3.2	2.1	20	0.563

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	34.9	16.2	32.9	16.2	32.0	16.1	18.8	28.4	18.1	27.9	17.4	27.5	4.1	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	13.9	21.3	15.1	12.9	21.0	14.0	12.0	20.6	60.0	28.6	57.5	27.9	55.2	27.1	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	9.5	8.0	-22.8	37.6	3.1	1.4	-25.0	38.5	-26.8	39.3	-28.5	40.1	-30.7	41.1		
(5)			-23.1	20.1	3.0	0.7	-25.3	20.6	-27.0	21.0	-28.7	21.5	-30.9	22.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	-1.2	-0.1	5.1	8.7	13.8	19.7	23.1	21.8	17.4	10.0	3.7	-1.5	(6)
(7)		DBStd	9.91	5.85	6.02	5.43	4.85	4.46	3.68	2.75	2.58	4.29	5.08	5.43	5.73	(7)
(8)		HDD10.0	1514	348	284	164	80	18	1	0	0	4	62	196	357	(8)
(9)		HDD18.3	3387	606	517	410	290	151	28	3	5	64	260	439	615	(9)
(10)		CDD10.0	1548	0	1	13	40	134	291	408	366	228	61	7	0	(10)
(11)		CDD18.3	379	0	0	0	0	8	68	152	113	37	1	0	0	(11)
(12)		CDH23.3	4979	0	0	6	29	224	974	1798	1300	583	65	1	0	(12)
(13)		CDH26.7	2187	0	0	0	2	63	424	899	570	219	10	0	0	(13)
(14)	Wind	WSAvg	3.2	2.8	3.2	3.6	4.0	3.4	3.2	3.0	2.9	3.0	2.9	2.8	(14)	
(15)	Precipitation	PrecAvg	414	11	16	32	55	71	43	42	39	42	30	18	15	(15)
(16)		PrecMax	559	39	38	100	167	198	96	98	93	209	67	42	58	(16)
(17)		PrecMin	219	1	4	0	5	23	0	9	5	1	5	4	5	(17)
(18)		PrecStd	87	9	9	26	34	39	26	22	24	39	18	10	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.2	18.8	23.8	26.3	31.5	36.4	37.4	35.2	33.9	27.8	22.2	15.9	(19)
(20)			MCWB	6.0	7.1	9.3	10.6	14.2	16.0	17.1	16.2	15.4	12.6	9.2	5.7	(20)
(21)		2%	DB	13.0	15.2	21.0	23.8	27.9	33.6	35.5	33.2	31.4	25.2	18.7	12.6	(21)
(22)			MCWB	4.4	5.5	8.3	10.0	13.3	15.5	16.4	16.3	14.9	11.5	7.9	4.3	(22)
(23)		5%	DB	10.9	12.5	17.8	21.9	26.1	31.3	33.7	32.2	29.1	22.4	16.2	9.9	(23)
(24)			MCWB	3.4	4.2	7.0	9.4	12.8	15.3	16.6	16.3	14.4	10.6	6.8	2.9	(24)
(25)		10%	DB	7.7	9.1	15.1	18.6	22.9	28.8	32.2	30.1	27.2	19.0	12.8	7.4	(25)
(26)			MCWB	1.7	2.5	5.8	8.4	12.0	15.0	16.7	16.1	14.0	9.4	5.1	1.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	7.5	10.1	12.2	16.6	19.1	20.1	19.3	18.1	13.6	9.7	6.2	(27)
(28)			MCDB	15.7	17.5	22.2	23.0	26.5	28.0	29.7	28.9	28.0	24.7	20.3	15.1	(28)
(29)		2%	WB	4.8	5.7	8.6	10.8	15.0	17.8	19.1	18.4	16.6	12.3	8.3	4.6	(29)
(30)			MCDB	12.7	14.7	19.4	21.7	24.1	27.8	28.8	27.9	26.8	22.6	17.9	12.1	(30)
(31)		5%	WB	3.3	4.3	7.4	9.9	13.9	16.8	18.4	17.7	15.6	11.1	6.9	3.0	(31)
(32)			MCDB	10.0	11.9	17.4	20.2	22.5	27.3	28.1	27.6	25.2	20.6	15.7	9.4	(32)
(33)		10%	WB	1.9	2.7	6.1	8.8	12.8	15.8	17.8	17.0	14.7	10.0	5.5	1.6	(33)
(34)			MCDB	7.5	9.0	14.6	17.8	21.7	26.2	27.7	26.8	24.1	18.4	12.6	7.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.2	13.2	14.8	14.2	14.2	15.8	15.5	15.4	14.2	13.5	12.7	(35)	
(36)			MCDBR	17.5	17.8	19.4	19.8	19.1	20.0	19.1	18.7	19.4	19.4	18.2	16.6	(36)
(37)			MCWBR	10.6	10.1	9.9	9.4	7.9	7.1	6.2	6.1	7.1	8.6	9.8	10.1	(37)
(38)		5% WB	MCDBR	16.5	17.0	18.6	18.0	16.2	16.9	15.6	15.7	16.3	17.0	17.7	15.9	(38)
(39)			MCWBR	10.3	9.9	9.6	8.9	7.4	6.9	5.8	5.9	6.5	8.0	9.7	10.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.244	0.253	0.277	0.304	0.327	0.334	0.364	0.357	0.315	0.277	0.258	0.248	(40)	
(41)		taud	2.596	2.579	2.522	2.472	2.464	2.461	2.420	2.429	2.532	2.615	2.582	2.579	(41)	
(42)		Ebn at Noon	952	986	986	971	948	938	905	905	930	943	922	918	(42)	
(43)		Edh at Noon	67	79	94	106	110	110	114	110	92	75	67	63	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.48	3.45	4.69	5.48	6.06	7.12	6.72	5.91	5.10	3.67	2.61	2.11	(44)	
(45)		RadStd	0.12	0.21	0.36	0.38	0.50	0.38	0.28	0.22	0.29	0.25	0.16	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.42	N/A	N/A	N/A	+0.55	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page